

Mahmoud Youssef

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EDUCATION

El Shorouk Academy

Bachelor of Science (B.Sc.) in Computer Science

Cairo, Egypt

Expected Jun. 2026

Harvard University (via edX / ITI)

CS50x — Introduction to Computer Science

Online

2024

PROJECTS

CalCounter — AI Nutrition Tracker | *Spring Boot 3, MySQL, JWT, HuggingFace, Docker*

[GitHub](#) · [Demo](#)

- Led backend development for a graduation-project team of 3 (backend, Flutter, AI); built 30+ endpoint RESTful API with JWT/RBAC via Spring Security, deployed on Render (Docker + MySQL) with Flutter web on Firebase Hosting
- Reduced food photo analysis wait time ~60% by integrating HuggingFace EfficientNetB3 Food-101 model via async pipeline (Spring @Async); beta-tested by 15+ users during graduation evaluation

Attendance Management API | *Spring Boot 3, MySQL, Redis, JWT, Docker*

[GitHub](#)

- QR-based workforce tracking with HMAC-SHA256 signed, time-boxed check-in sessions and Redis-backed per-employee deduplication, preventing duplicate check-ins while supporting team-wide sessions
- Schedule-driven attendance processing (PRESENT/LATE/HALF_DAY) with daily @Scheduled reconciliation that automatically marks missing records ABSENT and synchronizes approved leave ranges
- 34-endpoint REST API across 8 controllers with method-level RBAC (@PreAuthorize); reporting implemented using native SQL and recursive CTEs via EntityManager with zero N+1 queries

LogiBridge — B2B Order Management Platform | *Spring Boot 3, PostgreSQL, Redis, Docker*

[GitHub](#)

- Built 15+ endpoint order-management API; multi-role RBAC via Spring Security @PreAuthorize; JWT + refresh token rotation with reuse detection; immutable audit trail via @PreUpdate hook
- Exactly-once state transitions via PostgreSQL unique constraint + pessimistic locking with optimistic @Version fallback; idempotency keys (DB-backed) for safe network retries; validated with JUnit 5 + Mockito

E-Commerce Backend System | *Spring Boot 3, MySQL, Redis, Stripe, Docker*

[GitHub](#)

- 30+ endpoint REST API; decoupled order/payment flows via domain events (OrderCreatedEvent, PaymentCompletedEvent); applied SOLID + Clean Architecture across all modules
- Idempotent Stripe webhook (HMAC); two-phase inventory reservation + optimistic locking; Redis cache-aside cut read latency ~40%; SQL double-entry bookkeeping for revenue reporting

University Timetable Scheduler | *Spring Boot 3, MySQL, React, TypeScript, Docker*

[GitHub](#)

- Full-stack scheduling system; backend enforces 7 hard + 5 soft constraints with async generation and status polling; JWT + role-based access (ADMIN / SCHEDULER / INSTRUCTOR)
- React/TypeScript SPA with interactive weekly grid, multi-format export (PDF/Excel/PNG); CI/CD via GitHub Actions; Docker Compose deployment

Distributed URL Shortener | *Spring Boot 3, MySQL, Raw JDBC, Docker*

[GitHub](#)

- Redis INCR + Base62 distributed ID generation with zero collisions; AtomicLong fallback on Redis failure; atomic per-IP rate limiting via Redis; Raw JDBC for explicit query control
- Deterministic hash-based sharding (shortCode.hashCode() % N) routing requests across MySQL shards with no lookup table; cache-aside reads with DB-derived TTL and hourly @Scheduled cleanup of expired records

TECHNICAL SKILLS

Languages: Java (OOP, Collections, Streams, Lambda Expressions), SQL

Frameworks & Libraries: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, MapStruct, JdbcTemplate

Databases: MySQL, PostgreSQL, Redis (caching, rate limiting, session/TTL state)

Security: JWT Authentication, Refresh Token Rotation, RBAC, HMAC-SHA256 Signing

DevOps & Tools: Docker, Git, GitHub Actions CI/CD, Maven, Flyway, Swagger/OpenAPI, Postman, JUnit 5, Mockito

Methodologies: Object-Oriented Programming (OOP), SOLID, Design Patterns, Clean Architecture, RESTful API Design